Stage

An original spatial audio system for any song

Project plan, architecture and roadmap
Designed to fit Spotify's existing lossless delivery pipeline

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Stage is a working name. This is an independent proposal and is not affiliated with or endorsed by Spotify.
Interface mockups are illustrations of the idea, not Spotify designs.

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1 Executive summary

Spotify streams lossless audio, but it still plays every song as a plain stereo file. Apple Music, Tidal and Amazon Music offer spatial audio through Dolby Atmos, which gives listeners sound that surrounds them. That experience depends on a mix that was made specially for Atmos, so it exists only for the part of each catalog where artists and labels paid for that extra mix.

This document describes **Stage** (a working name; the final name is Spotify's to choose), an original spatial audio system designed to sit on top of Spotify's existing delivery pipeline. It takes the single stereo master that every artist already delivers, separates it into its musical parts, places those parts around the listener with a mix that is tuned to the song, and stores the result so that streaming it costs nothing extra at play time.

The five ideas that define the project

1. **Any song, not only specially mixed ones.** The engine works from the stereo master that is already in the catalog.
2. **No Dolby Atmos on the phone.** The spatial effect is rendered ahead of time as ordinary lossless audio, so any phone and any pair of headphones can play it.
3. **A mix that fits the song.** An Afrobeats record, a house record, a rap record and a pop record are each treated differently, and every track gets its own profile, even inside one album.
4. **Listening modes.** A studio-room mode, a car mode, a cinema mode and more, so the same song can be placed for the place where the listener actually is.
5. **Artists and labels stay in control.** A track gets a spatial mix only when its rights holder opts in, and nothing goes live without approval.

What makes it different from what exists today

Devices already offer generic stereo-to-spatial effects. They apply the same processing to every song and know nothing about the music or about what the artist wants. Stage is different in three ways: it works per track using separated parts, it follows rules that come from mixing practice rather than one fixed recipe, and it puts the rights holder in charge. It is also checked against the original mix, so the bass and the intent of the record are preserved.

> **How to read this plan**
>
> Stage is an **engineering and research project**, not a promise of a finished product that is better than Apple on day one. The goal of this plan is to build it as well as the available tools allow, to measure it honestly, and to show the result in a form that people at Spotify can hear for themselves.

In the rest of this document: the gap in the market and the product principles (chapters 2 and 3); how Spotify delivers audio today and where Stage plugs in (4 and 5); the architecture and each technical layer (6 to 10); artist control and the interface (11 and 12); data, tools and numbers (13 to 15); limitations with solutions, the roadmap and the pitch (16 to 18).

2 The problem and the opportunity

What spatial audio is

Ordinary stereo music has two channels, left and right. Spatial audio makes sounds appear at positions around you, at different directions and distances. On headphones this is done with **binaural rendering**, which filters each sound the way a real head and ears would, so the brain hears it as coming from a direction.

How Apple Music, Tidal and Amazon do it

Apple launched spatial audio with Dolby Atmos on Apple Music in June 2021. Dolby Atmos is an **object-based** format: the mix engineer places individual sounds as objects in a three-dimensional space and the player renders them for whatever the listener has. The catch is that this requires a new, separate mix for each song. Spatial audio therefore exists only for tracks where someone made an Atmos mix.

Playback also depends on the device. Apple's support page says an Android phone must support Dolby Atmos, and Windows needs the Dolby Access app. On iPhone, iPad and Mac, Atmos plays through any headphones because the device does the rendering.

Where Spotify stands

As of early 2026, reputable consumer guides still report that Spotify offers no Dolby Atmos or other native spatial format. Spotify did add lossless streaming in September 2025, which removed one of the main quality gaps with competitors. Spatial audio is now the most visible capability that Spotify lacks.

Listeners work around the gap with device-level effects such as Apple's "Spatialize Stereo" or Windows Sonic. These are generic: one setting for every song, no knowledge of the music, and no say for the artist.

The opportunity

	Dolby Atmos catalogs	Device stereo upmix	Stage (this plan)
Needs a special mix per song	Yes	No	No
Works on the whole catalog	No, only Atmos mixes	Yes	Yes, for opted-in tracks
Tuned to the song	Yes, by an engineer	No	Yes, per track
Artist or label control	Yes, they make the mix	None	Yes, opt-in and approval
Depends on the phone's Dolby support	Yes	Varies by device	No, rendered in advance

Confirmed: the device dependency is real

Tested on an Android 16 Samsung Galaxy phone: with the system Dolby Atmos switch on, an Apple Music Atmos track plays spatially; with it off, the track plays and the Atmos label stays, but the sound is not spatial. Stage removes this dependency, because Spotify would own the rendering instead of relying on each phone.

3 Vision and product principles

Vision

Any song on Spotify, if its artist agrees, can be heard as if it were mixed for a space around you, in a form that suits where you are listening, with no special hardware and no loss of quality.

Principles

These principles come directly from what you asked for. They are written as rules that every later design decision has to respect.

Principle	What it means in practice
True to the record	The engine places and shapes sound; it never rewrites the performance. Levels, tone and the low end stay faithful to the original master. A built-in check compares the spatial render with the original and flags drift.
Not generative	No new notes, no replaced instruments. The model separates and arranges existing audio only.
Fits the song	Style is detected for each track. Two songs on one album can receive different profiles.
Depth over gimmick	Width and movement are used where they serve the music. Anchor sounds such as kick, bass and lead vocal stay solid.
Lossless all the way	Output is 24-bit lossless audio at the platform's top supported quality, with no extra lossy step.
Computed ahead of time	The heavy work happens once per track in an offline job. Playback is a normal file stream.
Rights holder in control	Opt-in per track, preview before release, and the ability to turn it off.
Listener in control	Spatial is a toggle. The normal stereo mix is always one tap away.
Honest about limits	Every weakness is measured and documented, not hidden.

Who it is for

- **Listeners** who want a more immersive experience without buying new hardware or switching service.
- **Artists and labels** who want their catalog to be heard spatially without paying for a new mix of every back-catalog song, and who want to approve how it sounds.
- **Spotify**, which gains a feature that its main competitors already promote, built on audio it already holds.

4 How Spotify works today

To design something that plugs into Spotify, you need to know what Spotify already does. This chapter summarises what is publicly known. Most of it comes from Spotify's own help pages and from published engineering write-ups. Where a detail comes from third-party summaries, it is marked as approximate.

What artists and labels deliver

- Spotify says it strongly prefers **FLAC**, and it also accepts WAV that meets the same technical rules.
- The delivery must be the native stereo master: 44.1 kHz or higher, 24-bit if that is how it was mastered, 16-bit only if nothing better exists, and two channels.
- Spotify supports **one stereo master per track**. Artists are told not to downsample or reduce bit depth themselves.
- Spotify converts WAV to FLAC internally and handles all other format work. Material above 44.1 kHz or 24-bit is reduced internally to a maximum of 44.1 kHz and 24-bit FLAC.

> **Why it matters**
>
> This is the single most important fact for the project. Because the delivery contains only a stereo master, there are no stems and no multichannel files to work from. Stage therefore has to start from stereo, which is exactly what the plan assumes.

What listeners receive

- Lossless streaming arrived for Premium users in September 2025: FLAC at up to 24-bit and 44.1 kHz, at no extra price, rolling out across more than 50 markets.
- Before that, the top quality was lossy (Ogg Vorbis, with a 320 kbps ceiling on the highest setting).
- Lossless is switched on manually on each device. Bluetooth cannot carry full lossless because of bandwidth, so Spotify recommends wired headphones or Wi-Fi playback through Spotify Connect.
- Spotify said nearly every track in its library supports lossless.

The backend, in outline

The following is a summary from public engineering write-ups and industry articles. Details change over time, so treat it as a map rather than a specification.

Area	What is publicly described
Architecture	A large set of microservices. Spotify was an early, well-known adopter of the approach.
Languages	Mainly Java, with Scala, Python and others in use.
Cloud	Spotify moved from its own data centers to Google Cloud around 2016.
Orchestration	Kubernetes runs the containers.
Events and data	Apache Kafka carries play, skip and other events. Cassandra stores large distributed data sets. BigQuery and data pipelines support analytics.
Machine learning	TensorFlow and PyTorch are cited for model work, with workflow tooling around them.

Area	What is publicly described
Delivery	Audio files are fetched from a central store through edge servers and cached close to the listener and on the device.

The streaming path

1. The app asks for a track. The request goes through Spotify's backend, which checks the account, rights and the quality setting.
2. The app fetches the audio file in pieces. The device cache is checked first, then nearby edge servers, then the central store.
3. The app decodes the stream and plays it. Spotify Connect lets other devices play the stream directly.
4. Playback events are sent back through the event system for recommendations, payments to rights holders and analytics.

What this means for Stage

- Stage does not need to change how audio is stored, cached or delivered. A spatial render is just another audio asset attached to a track.
- The existing quality setting and the Connect picker already show how a quality badge is presented in the app, which is a model for a spatial badge.
- The two things Spotify has not built are the processing engine and the artist consent flow. Those are what this plan designs.

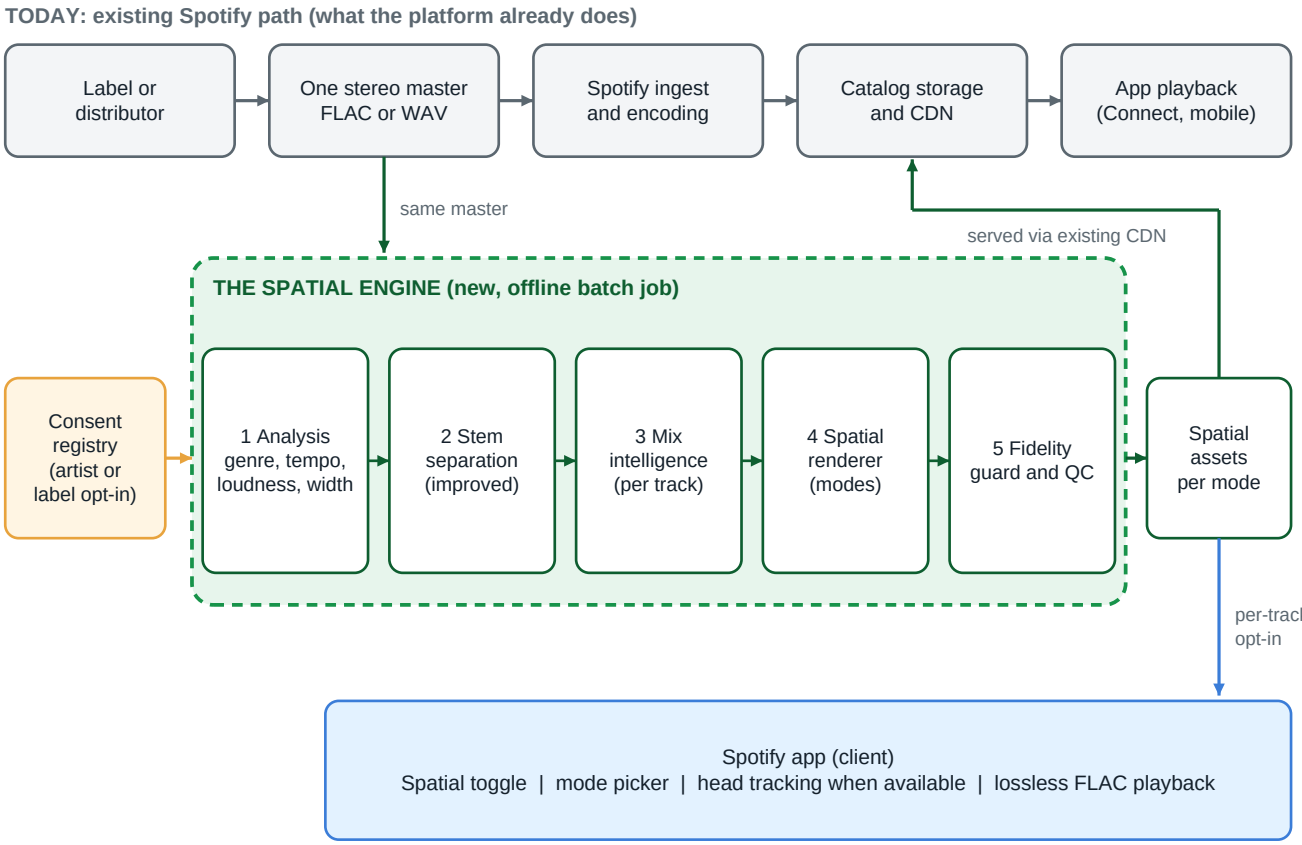


Figure 1. Stage as an add-on. The top row is how Spotify works today; everything below it is new.

5 Where Stage fits into Spotify

Integration points

A feature is easiest to adopt when it needs few changes to what already exists. Stage touches Spotify in exactly four places.

Point	What Stage adds	Why it is a small change
Ingest	A hook that, for opted-in tracks, sends the stereo master to the spatial engine after normal ingest.	The master is already there. No new delivery format is needed from artists.
Rights and consent	A per-track setting that records the rights holder's decision and approval.	It is metadata attached to an existing track record.
Storage and CDN	One or more extra audio assets linked to a track, stored and cached like any other file.	FLAC is already the supported format.
Client app	A spatial toggle, a mode picker, and a badge in the Now Playing view.	It follows the pattern of the existing lossless indicator.

Two ways to deliver the result

There are two sensible designs, and the choice shapes the storage bill and the quality of the experience.

	Option A: pre-rendered per mode	Option B: stems plus placement data
What is stored	A finished stereo FLAC file for each mode.	Separated stems and a small file of placement instructions.
Playback	Identical to any normal track. The phone only decodes.	The phone renders the spatial effect live from the stems.
Storage	Grows with each mode offered.	Larger up front, but the same for every mode.
Phone requirements	None.	Needs enough CPU for real-time rendering.
Head tracking	Hard. The effect is baked in.	Natural. The renderer can react to head movement.
Risk to the original	Original stays untouched; spatial is a separate asset.	Stems are separation outputs, which carry artifacts.
Best for	The first release and the demo.	A later generation of the product.

The plan starts with **Option A** because it needs no new playback code on the phone and keeps lossless quality simple. The architecture is built so that Option B can be added later without changing the analysis, separation or mix-intelligence layers.

The demo environment

You do not need Spotify's servers to prove the idea. The demo is a self-contained system: the engine runs on your laptop, outputs are stored as files, and a listener app (web or Android) plays them with a spatial toggle and a mode picker. It mimics each integration point above so that the path to a real deployment is clear.

6 System architecture

Stage is a pipeline of five stages that run offline on each opted-in track, followed by storage and a client. Each stage has a clear input, a clear output, and a way to test it on its own. This separation is deliberate: it lets you improve one layer without breaking the others.

The pipeline

Stage	Input	Output	Purpose
1 Analysis	Stereo master	Feature report	Measures tempo, key, loudness, stereo width, dynamics, spectral balance and estimates style and era.
2 Stem separation	Stereo master	Stems and a residual	Splits the song into vocals, drums, bass and other parts, plus a leftover signal so nothing is lost.
3 Mix intelligence	Stems and feature report	Placement plan	Decides where each part sits, how wide and how far, and what room character fits.
4 Spatial renderer	Stems and placement plan	Spatial stereo per mode	Turns the plan into audio using head-related filters and a room model.
5 Fidelity guard	Original and spatial render	Pass, fail, report	Checks the result against the original and blocks anything that drifts too far.

Design rules for the whole system

- **Stems are never the only copy of the sound.** The renderer always blends separated stems with a residual so that the sum equals the original mix before any placement is applied.
- **Every decision is logged.** The placement plan is a small, readable file. This makes each mix explainable, repeatable and easy to hand to an artist for approval.
- **Deterministic where possible.** The same track and the same settings produce the same output. This matters for testing and for trust.
- **Fail safe.** If any stage fails or the guard rejects the result, the track simply stays in normal stereo. A failed render is never shown to a listener.
- **Modular models.** The separator, the style classifier and the placement policy are each replaceable components behind a stable interface.

Job orchestration

When a rights holder opts in and a track is ingested, a job is created. The job moves through the stages and writes its result and its logs next to the track. At Spotify's scale this would be a queue with many workers. On your laptop it can be a simple local queue that processes one track at a time. The interface is the same, so the demo shows the real design.

The one-to-two-hour promise

You asked that a spatial mix be ready within one to two hours of a track being released. That is realistic for a batch system, because nothing has to happen in real time. The budget has to cover separation, analysis, planning, rendering for each mode, quality checks, and a short human approval window if the artist wants one. Chapter 15 shows the numbers.

Storage layout for one track

Item	Description
original	The stereo master as delivered. Never modified.
stems	Separated parts, kept for re-rendering and for the later stems-plus-data design.
analysis.json	The feature report from stage 1.
plan.json	The placement plan from stage 3, in readable form.
render_[mode].flac	One finished lossless file for each mode.
qc.json	The fidelity guard report with the measured differences from the original.
consent.json	Who opted in, when, under which agreement, and any approval.

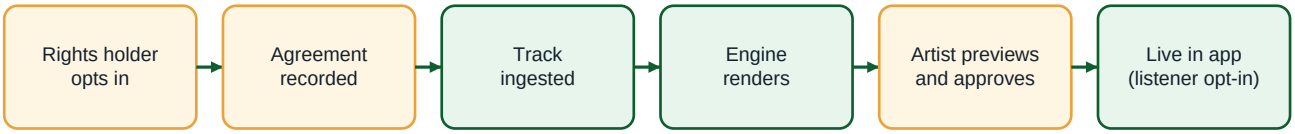


Figure 2. The life of a track. Amber steps involve the rights holder; green steps are automatic.

7 Layer 1: stem separation

Stem separation is the foundation. If the parts are dirty, everything built on top of them suffers. You want separation that is much cleaner than the typical consumer tools, so this chapter treats it as a research problem.

What separation does and why it is hard

A stereo master is a sum of many sounds. Separation tries to recover each sound from the sum. The challenge is that sounds overlap in time and in frequency, effects such as reverb and compression blur their edges, and the original separate recordings are not available to compare against. Most systems learn to estimate each part from examples of songs where the true stems are known.

The starting point: Demucs

Demucs is a family of open-source separation models developed at Meta AI. Its current form, **Hybrid Transformer Demucs**, works on both the raw waveform and the spectrogram and combines them with a transformer, and it splits music into drums, bass, vocals and other, with a larger variant that also tries guitar and piano. It is the standard open baseline. Before building on it, check the current state of the repository and the license of both the code and the pretrained weights.

How to go beyond the baseline

You are right that the baseline can be improved. These are the realistic routes, in order of how much each is expected to help for the effort.

Technique	Idea	Expected benefit	Cost
Ensembling	Run several models or settings and average the outputs, often weighted per stem.	Reduces random errors and artifacts.	More compute per track.
Test-time augmentation	Separate shifted and flipped copies of the audio and average the results.	Smoother output, fewer glitches.	Several times more compute.
Residual refinement	A second small model that looks at each stem, the mix and the leftover and removes bleed.	Targets exactly the leakage you described.	Needs training data and design effort.
Fine-tuning for your music	Continue training on material close to your target genres, including African popular music.	Large gains where the baseline saw little similar music.	Needs legal training data and GPU time.
Mixture-consistent output	Force the stems and the residual to sum exactly to the input.	Guarantees nothing is lost or invented.	Low.
Genre-aware weighting	Choose model settings per detected style.	Small to moderate.	Low once classification exists.

Why a residual matters

A separator never recovers parts perfectly. If you throw away what it did not assign, you lose real audio, and the song sounds thinner. Keeping a residual (the original minus the sum of the stems) means that, before placement, recombining everything restores the exact original. This also supports your rule that the engine must not remove anything from the record.

Measuring separation honestly

- **SDR, SIR and SAR**: standard scores for overall quality, interference from other sources, and artifacts. They are computed against known stems.
- **SI-SDR**: a scale-invariant variant often used in training.
- **Listening tests**: numbers do not catch everything, so you also run blind comparisons on difficult passages such as sibilant vocals, reverb tails and dense percussion.
- **Task-specific tests**: because the stems are used for placement, check how artifacts behave after placement, not only in isolation.

Data for training and testing

- **MUSDB18-HQ**: 150 songs with four stems each. It is the common benchmark, but it is small and mostly Western pop and rock.
- **MoisesDB**: a larger collection with finer-grained stems.
- Datasets are usually licensed for research use, so check each license before training anything you plan to show or release.
- **Your own material**: songs where you hold the multitrack sessions (your productions, or collaborators who agree) are the safest source of both training data and public demos.

> **On improving it for free**
>
> You can improve a strong open model without paying for anything, but expect the gain to come from careful engineering, good data and honest evaluation, not from a single trick. Free GPU tiers from providers such as Kaggle or Colab exist, though their limits change and sessions are short, so plan training in small, resumable runs.

The practical risk

Even excellent separation leaves small errors. Chapter 8 handles this by design: sounds that expose artifacts badly (a lead vocal, a kick) stay near the centre and keep a strong share of the original, while the elements that hide artifacts well (reverb, pads, percussion, backing vocals) carry most of the width and movement.

8 Layer 2: mix intelligence

This is the part that is original to the project and the part that makes it more than a wrapper around an existing separator. It decides, for each track, where every element goes and what the space around it feels like.

What the system looks at

Signal	How it is obtained	How it is used
Style and era	A classifier on audio features, plus metadata if available.	Chooses the starting template: Afrobeats, amapiano, house, hip-hop, pop, R&B, rock and others.
Tempo and groove	Beat tracking and onset analysis.	Sets timing for any movement and for reverb length.
Stem energy and role	Level, spectral shape and activity of each stem over time.	Decides which stem is the lead, which are supporting, and which are texture.
Stereo width of the original	Mid/side analysis of the master and of each stem.	Keeps the spatial result consistent with how wide the original already was.
Dynamics and loudness	Short-term and integrated loudness.	Keeps the spatial version at a matching level and avoids over-compression.
Frequency balance	Spectral analysis per stem.	Protects the low end and flags sounds that would clash if moved.
Era and recording type	Mono or stereo detection, bandwidth, noise floor.	Older mono recordings get a gentler image than modern wide mixes.

The placement plan

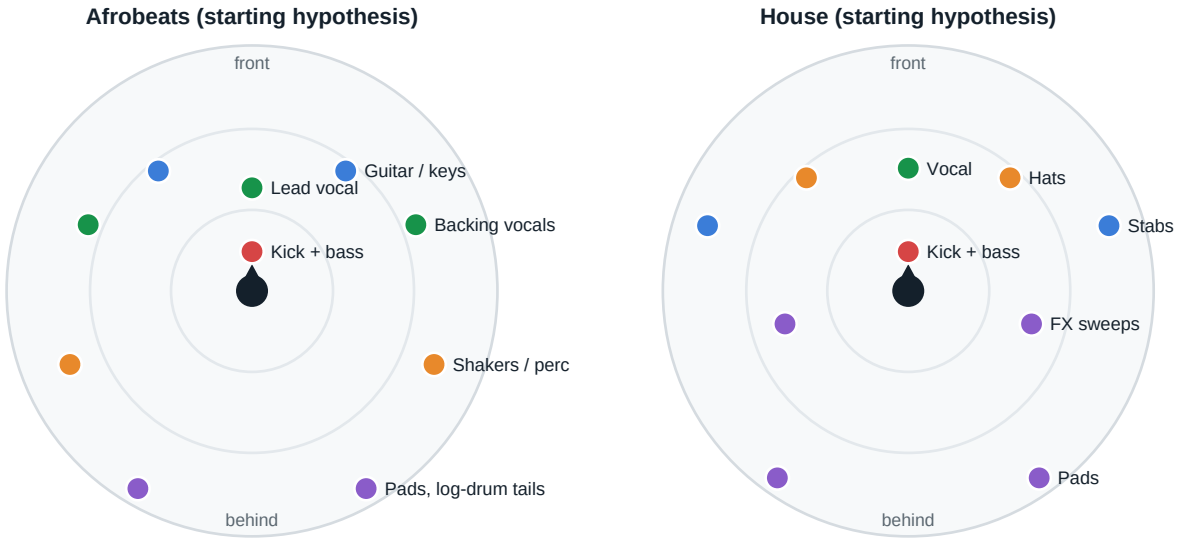
For each stem, and sometimes for parts of a stem, the plan specifies a handful of numbers.

Parameter	Meaning
Azimuth	The left-right angle around the listener, from 180 degrees on the left, through 0 in front, to 180 on the right.
Elevation	How high or low the source appears.
Distance	How near or far. Nearer sounds are louder, drier and clearer; farther ones are softer and more reverberant.
Spread	How wide the source is, from a point to a broad image.
Room send	How much of the source feeds the room model and early reflections.
Movement	Optional slow motion, used sparingly and only on elements that tolerate it.
Original blend	How much of the original, unmoved signal stays in the mix for this element.

Rules that come from mixing practice

The first version of the policy is **rule-based**, written from established practice in stereo and immersive mixing. These rules are starting hypotheses. They are meant to be tested by listening and revised.

- **Anchor the foundation.** Kick, bass and the lead vocal stay at or near the centre. Low frequencies are kept mono, because they are hard to localise and because phones and car speakers handle them better that way.
- **Put width where it is safe.** Reverb, pads, backing vocals, shakers, hats and ambience carry the width. If separation left small artifacts, they are least audible here.
- **Respect the mix balance.** Placement changes position, not loudness relationships. A sound that was prominent stays prominent.
- **Avoid constant motion.** Movement is used as an accent, not a default. Too much motion tires listeners and sounds like an effect.
- **Keep the original as a safety net.** A share of the original mix is blended back so that any separation error is masked by the real recording.
- **Do not change the groove.** No timing changes, no re-tuning, no filtering of the performance.



Top-down view from above the listener. Dots mirror left and right. Distance from the head is perceived depth.

Figure 3. The same idea applied differently. The layouts are starting hypotheses to be tested by ear, not fixed rules.

Genre templates as starting points

Style	Features to expect	Starting approach (to be validated)
Afrobeats and Afropop	Layered percussion, shakers, log drums, call-and-response vocals, strong groove.	Centre the kick, bass and lead. Spread percussion and backing vocals wide. Give log-drum tails and pads depth behind the listener.
Amapiano	Deep log-drum bass, sparse top end, long atmospheric elements.	Protect the sub bass in mono. Use depth and space on pads and vocal chops. Keep the groove tight.
House and dance	Steady kick, driving bass, repeating stabs, sweeps and builds.	Centre kick and bass. Place stabs and hats wide. Use FX sweeps for controlled movement during builds.
Hip-hop and rap	Dominant vocal, heavy drums, sparse arrangement.	Keep the vocal forward and clear. Use width for ad-libs, samples and room. Avoid pushing the drums away from the centre.
Pop	Polished vocal, layered production, big chorus.	Lead vocal front and close. Open the image on choruses. Use stacked vocals for width.
R&B and soul	Intimate vocals, harmonies, warm instrumentation.	Closer, warmer space. Harmonies around the lead. Gentle reverb depth.
Rock and band music	Guitars, live drums, bass, vocal.	Drums as a stage in front, guitars left and right, vocal centre. Natural room sound.
Older and mono recordings	Narrow image, limited bandwidth.	A restrained, believable space. Little movement. Avoid pretending the recording was wide.

From rules to learning

A rule-based policy is the right first step because it is explainable and gives you a reliable baseline. The plan then grows toward learned behaviour in three steps.

1. **Per-track tuning.** Parameters inside each template adapt to measured features of the track, such as how dense the percussion is or how wide the original already was.
2. **Learned style similarity.** A model finds the closest examples among tracks that were mixed well and borrows their placement shape, using any properly licensed Atmos or multichannel references you can obtain.
3. **Preference learning.** Listening-test results become training signals, so the system learns which choices people prefer for each style.

Because this is a design built to be measured, each step has to beat the previous one in blind listening tests before it replaces it.

The fidelity guard

Your requirement that the engine not alter the record becomes a set of automated checks that run on every render.

Check	What it tests	Why
Downmix comparison	Fold the spatial render back to stereo and compare with the original in level and spectrum.	Large differences mean the engine changed the record's character.
Low-end integrity	Compare energy and phase below roughly 120 Hz.	Protects the bass you asked it not to touch.

Check	What it tests	Why
Loudness match	Compare integrated loudness against the original.	Prevents the spatial version from sounding louder or quieter.
True peak and clipping	Check peaks after rendering.	Avoids distortion.
Mono compatibility	Sum to mono and check for cancellation.	Phones, car stereos and clubs often play mono.
Artifact scan	Look for glitches near stem boundaries and abrupt level changes.	Catches separation errors before listeners do.

> **Thresholds are tuned, not guessed**
>
> Any limit in the guard (for example, how many decibels of deviation is allowed in the downmix) is a tunable setting. The right values will be found by listening to real tracks, and they may differ by style.

9 Layer 3: the spatial renderer

The renderer turns a placement plan into audio. For headphones, it is a well-understood technique: binaural synthesis. This chapter explains how it works and the mathematics involved, in plain language first and then with the key formulas.

How people locate sound

- **Interaural time difference (ITD)**: a sound from the right reaches the right ear slightly earlier than the left.
- **Interaural level difference (ILD)**: the head shadows high frequencies, so the far ear hears them quieter.
- **Spectral cues**: the shape of the outer ear filters sound differently depending on direction, which is how we tell front from back and up from down.
- **Distance cues**: louder, drier and brighter sounds are heard as closer. Quieter, more reverberant and duller ones are heard as farther.
- **Head movement**: small movements resolve ambiguity. This is why head tracking makes spatial audio more convincing.

The core idea: head-related transfer functions

A head-related transfer function (HRTF) describes how a sound from a given direction is changed on its way to each eardrum. In the time domain this is a short filter called a head-related impulse response (HRIR). To place a sound at a direction, you filter the sound with the HRIR for the left ear and the right ear.

```
left(t) = source(t) * HRIR_left(azimuth, elevation)
right(t) = source(t) * HRIR_right(azimuth, elevation)
(the * is convolution; each source gets its own pair of filters)
```

Two cues are easy to model on their own and give intuition. The classic Woodworth model estimates the time difference for a spherical head of radius a at azimuth angle theta:

```
ITD(theta) = (a / c) * (theta + sin(theta))
a is about 0.0875 m (head radius), c = 343 m/s (speed of sound)
At 90 degrees this gives about 0.66 milliseconds, the largest delay a human head
produces.
```

Distance is modelled with the inverse-distance law. Each doubling of distance reduces the direct sound by about 6 dB, while reverb stays roughly constant, so the balance between direct and reverberant sound signals distance:

```
level_dB(d) = level_dB(1 m) - 20 * log10(d / 1 m)
```

Where the HRTFs come from

- **Measured datasets.** Free public datasets store HRTFs from real people and dummy heads in the standard SOFA format (AES69). Examples include CIPIC, HUTUBS and SADIE II. Check each license.
- **Generic versus personal.** A generic HRTF works acceptably for many listeners and badly for some. This is a known limitation of all binaural systems.
- **Personalisation.** Later versions can let a listener choose among a few HRTF profiles or estimate a fit from simple measurements. Spotify could do this in the app; the demo can offer a short selector.

Rooms and reverb

A believable space needs a room around the sources. The renderer builds it in two parts.

- **Early reflections**: the first few echoes from walls, computed with a simple geometric method (the image-source method). They tell the brain how big and how close the room is.
- **Late reverb**: the dense tail. This can be synthesised with a feedback delay network, or taken from a measured impulse response of a real room.
- **Matching the music**: reverb time and brightness are set from the style and the tempo so that the room supports the groove instead of smearing it.

Doing it efficiently

Convolution with long filters is expensive, so the renderer uses fast Fourier transforms and partitioned convolution, which splits long filters into blocks. Because the work is offline, speed is not critical, which lets you favour accuracy. Doing it offline also means you can render at higher internal precision and only convert to the output format at the end.

Loudness, bass and output

- **Bass management**: low frequencies are summed to the centre before rendering so they stay solid and mono-compatible.
- **Loudness**: the output is matched to the original using the ITU-R BS.1770 loudness measure (the basis of LUFS), because Spotify applies its own loudness normalisation and a mismatch would alter how the spatial version is perceived.
- **Limiting**: a true-peak limiter prevents clipping after rendering.
- **Format**: the final file is 24-bit lossless FLAC, the same family Spotify already streams.

Head tracking

Head tracking keeps sounds fixed in space as you turn your head. It needs a motion sensor in the headphones and a renderer that updates in real time, so it belongs to the stems-plus-data design (Option B). Platforms expose head orientation to apps on some devices, and support depends on the phone and headphones. The architecture reserves a place for it but the first release does not require it.

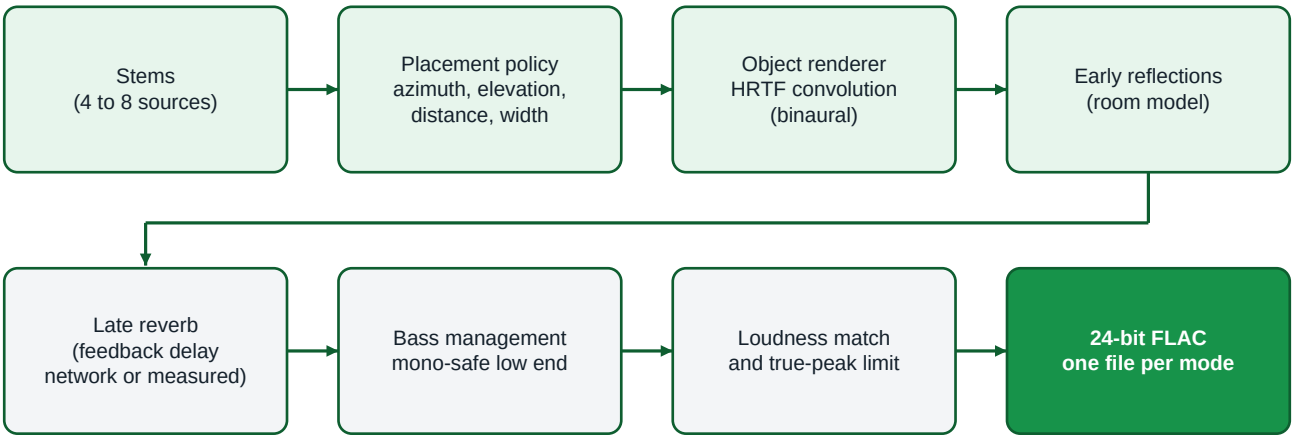


Figure 4. The rendering chain for one mode, from separated stems to a finished lossless file.

Speakers and cars

Speakers are harder than headphones because each ear hears both speakers. Two techniques apply.

- **Crosstalk cancellation**: sends a small inverted signal from each speaker to cancel the unwanted path to the opposite ear. It works well only when the listener sits in a known position.
- **Cabin modelling**: for a car, a response of the cabin (reflections, speaker positions and the effect of seat position) is modelled and the mix is adjusted for it.

This is why headphone modes come first and car mode is treated as its own research task, with clear honesty about how well it generalises across different cars.

10 Listening modes

Modes are how Stage adapts one song to different places. Each mode is a bundle of settings: a room model, a placement style, a target playback system and a loudness and bass treatment.

The mode set

Mode	Target playback	Character	Main technique
Studio Room	Headphones	Close, warm, accurate. Like sitting in a well-treated control room.	Binaural, short early reflections, subtle room.
Cinema	Headphones or speakers	Wide and large, with a sense of a big screen in front.	Binaural with a larger room and stronger front image.
Club	Headphones or speakers	Punchy and energetic, controlled low end, a sense of a crowd space.	Tight bass management, brighter reflections.
Open Air	Headphones	Natural and airy with little reverb, like an outdoor stage.	Minimal reflections, distance cues.
Car	Car speakers	Adjusted for cabin acoustics and seat position.	Speaker rendering and cabin correction.
Stadium or Arena	Headphones	Very large space, long decay, a sense of scale.	Long reverb, strong distance cues.

About naming modes after famous studios

You suggested using names of famous studios, for example Westlake, as mode names. It is a good idea from a storytelling point of view, but it has two issues to plan around.

- **Trademarks.** Studio names are trademarks of their owners. A public demo should use original names, or reach written permission, before using a real studio's name.
- **Accuracy.** To honestly claim a mode sounds like a particular room, you need measured impulse responses from that room, which are generally not public. Without them, the mode is a model of a type of room, not that room.

A safe approach is to give each mode an original name and a plain description, and to say "inspired by the feel of a classic tracking room" in the write-up. If Spotify later partners with a studio, measured responses from that room could become a premium mode.

How modes interact with the mix intelligence

Modes and mix intelligence are separate layers. The mix intelligence decides where each element sits relative to the others. The mode decides the space those elements sit in. The same placement plan can therefore be rendered for several modes, which keeps the cost of adding a mode low.

Automatic mode suggestions

The app can suggest a mode using what it already knows: when audio is routed to a car through Bluetooth or a wired connection, Car mode can be offered. When headphones are detected, Studio Room can be the default. The listener is always free to choose.

11 Artist and label control

A system that remixes people's music, even gently, has to earn their trust. This is both a product principle and a practical requirement: without rights holders' consent, a feature like this cannot ship.

The control model

Level	Who decides	What they decide
Catalog	Label or distributor	A default for the whole catalog: off, opt-in per track, or on for all new releases.
Release	Artist or label	Whether an album or single gets spatial mixes.
Track	Artist or label	Whether this song gets one, and which modes are offered.
Listener	The person listening	Whether to play the spatial version. Stereo is always available.

The workflow

1. A signed agreement between the rights holder and Spotify authorises spatial processing for the chosen tracks.
2. The rights holder switches spatial mixing on for a track or a release in a Spotify for Artists page (Figure 5, right).
3. The engine renders the track. The rights holder can listen to a preview of each mode.
4. They approve, request a different intensity, or decline. Nothing goes live without approval if the label chose that setting.
5. Once approved, the track becomes available to listeners, who choose whether to use it.
6. The rights holder can withdraw at any time, and the spatial asset is removed.

Controls an artist might want

- **Intensity.** A simple slider from subtle to bold, which scales width, depth and movement.
- **Protected elements.** A rule such as "keep the lead vocal centred" or "no movement on the drums".
- **Exclusions.** Sections or tracks that should never be processed, for example a recording where the stereo image is part of the art.

Legal and ethical points (product design, not legal advice)

- **Derivative works.** A processed version of a recording may count as a new version for rights purposes. This is why a signed agreement is part of the design.
- **Moral rights and artistic intent.** Some artists will not want their records altered. Opt-in respects this, and the ability to decline must be easy.
- **Labelling.** The listener-facing name should not imply that the artist or an engineer personally made the mix unless that is true.

12 User experience and interface

Your stated aim is to improve the experience. The interface therefore needs to feel like a natural part of Spotify, stay out of the way, and make the benefit obvious in a few seconds.

Design principles

- **One tap to try.** A visible but quiet spatial control in Now Playing, next to the existing quality indicators.
- **Honest feedback.** A clear badge shows when spatial is on and which mode is active.
- **Instant comparison.** Switching between stereo and spatial keeps the playback position, so the listener can hear the difference without losing their place.
- **Few controls.** A mode picker and two simple sliders, depth and width. Advanced settings are available but hidden.
- **Respect the platform.** Use Spotify's existing patterns, such as bottom sheets, toggles and badges.

Key screens

The mockups below are illustrations of the idea, using a dark theme similar to Spotify's. They are not Spotify designs.

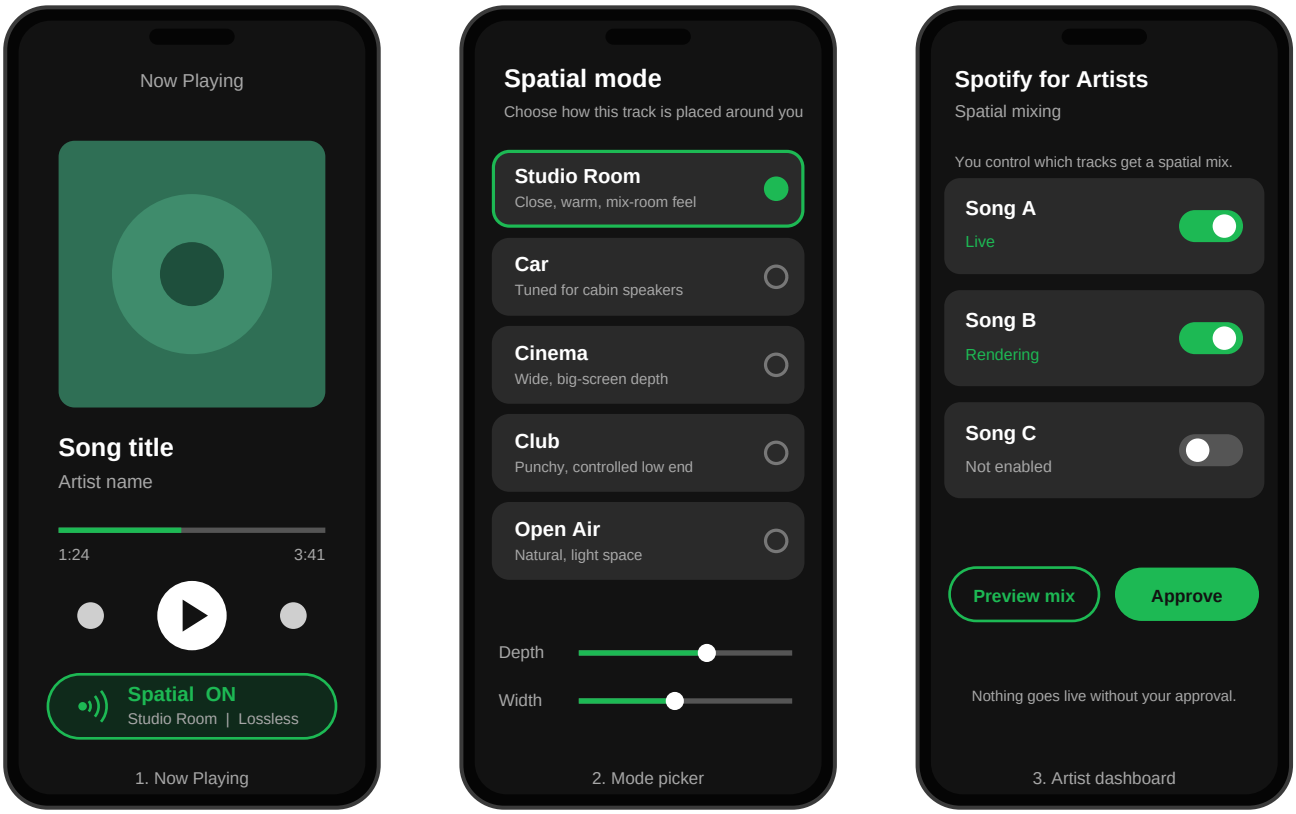


Figure 5. Left: Now Playing with a Spatial badge and the lossless indicator. Centre: mode picker. Right: the artist dashboard.

User flows

Flow	Steps
First use	A short one-time card in Now Playing: "Hear this song around you." Tap, the mode picker opens with Studio Room selected, and the track plays in spatial.
Switch mode	Tap the badge, choose a mode. Playback continues without a gap. The change is applied at a safe point in the track.
Compare	Long-press the badge to flip between stereo and spatial for the same moment in the song.
Car connection	On connecting to a car, a prompt offers Car mode. Declining remembers the choice.
Unavailable	If a track has no spatial version, the control shows why ("The artist has not enabled spatial mixing for this track") instead of silently doing nothing.
Headphone fit	In settings, a short selector lets the listener pick the HRTF profile that sounds most natural.

Accessibility and inclusion

- Every control is labelled for screen readers, and the toggle state is announced.
- The default must be safe for listeners who dislike effects: spatial is off until they choose it.
- Some listeners experience discomfort with strong spatial effects. An intensity slider and an easy off switch address this.
- Data use is unchanged from lossless playback, and an option to limit spatial downloads protects people with expensive data, which is a real concern for many listeners worldwide.

Content of the artist dashboard

- A list of tracks with spatial status: not enabled, rendering, ready for approval, live.
- Preview players for each mode with a simple switch between original and spatial.
- Intensity and protected-element controls, and approve or decline buttons.
- Plain-language information on what the processing does and does not change.

13 Data, testing and evaluation

Building the system and proving that it is good are two jobs. This chapter covers the data you need and the tests that will tell you, and later others, whether the result is actually better.

The test library

You described testing across the decades, from the 1950s to the 2020s, and across genres. That is exactly right, and it should be organised deliberately.

Dimension	Coverage to aim for
Era	1950s and 1960s (often mono or early stereo), 1970s to 1990s, 2000s, 2010s, 2020s.
Genre	Afrobeats, highlife, amapiano, hip-hop, pop, R&B, house, rock, soul, jazz, classical, reggae, gospel.
Mix density	Sparse tracks, dense tracks, vocal-led, beat-led, live recordings.
Recording type	True mono, narrow stereo, wide stereo, heavily processed.
Difficulty	Sibilant vocals, heavy reverb, distorted guitars, sub-bass, fast percussion.

> **Develop with anything, publish with the right material**
>
> You can *listen* to commercial records on your own while developing. For anything you publish, such as a demo video, use music you own, music with an explicit open license, or tracks from artists who gave permission. Public posts that contain other people's records are likely to be removed and could damage the project's credibility.

Objective metrics

Metric	What it measures	Used for
SDR, SIR, SAR	Separation quality, interference, artifacts.	Stem separation (needs reference stems).
Downmix difference	Spectral and level difference after folding to stereo.	Fidelity to the original.
Low-end correlation	Similarity of the bass region.	Bass preservation.
Loudness difference	Integrated and short-term loudness against the original.	Level consistency.
Interaural measures	Time and level differences produced for each placed source.	Checking the renderer against its own plan.
Inter-channel correlation	Width and mono compatibility.	Spatial image health.

Listening tests

Numbers cannot tell you whether a mix is enjoyable. Listening tests are the real test, and they can be run for free.

- **A/B and ABX**: blind comparisons between the original and the spatial version, or between two versions of the system.
- **MUSHRA-style ratings**: listeners rate several versions of the same clip on a scale, with a hidden reference and a low-quality anchor.
- **Preference tests by style**: ask whether listeners prefer the spatial version, and why.
- **Panel**: friends, classmates, the IEEE chapter and online communities of musicians and engineers. Include trained listeners, such as producers, and ordinary listeners, as they judge differently.
- **Controls**: use the same loudness for all versions, randomise order, use good headphones, and record the headphone model.

Success criteria you can state in advance

Question	Evidence that would count as success
Is the separation clean?	Higher SDR than the baseline on a held-out set, and listeners rate difficult passages as cleaner.
Is the mix true to the record?	Fidelity guard passes on the full test library, and listeners do not report changes to bass or intent.
Does it sound better than generic upmixing?	Listeners prefer Stage over a generic stereo upmix in blind tests.
Does style matter?	Style-aware profiles are preferred over a single fixed profile.
How close is it to a human Atmos mix?	Report the gap honestly, for the tracks where an Atmos reference can be compared.

Writing these criteria before testing protects the project from wishful thinking, and it also makes the results much more convincing to anyone who reads the write-up.

14 Technology stack

Every tool below is free to use. The point of this chapter is to show the options for each layer and to recommend one.

Recommended stack by layer

Layer	Recommended	Alternatives	Notes
Language for research	Python 3	Julia	Largest ecosystem for audio and machine learning.
Deep learning	PyTorch	TensorFlow, JAX	Demucs is in PyTorch.
Separation baseline	Demucs models	Open-Unmix, other open models	Check licenses and maintenance status.
Audio analysis	librosa, torchaudio, essentia (check license)	madmom, aubio	Beats, onsets, spectra, key.
Style classification	Small CNN or audio embeddings on an open dataset	Pretrained audio models	Needs a labelled genre set including African styles.
Signal processing	NumPy, SciPy	JAX	Convolution, filtering, resampling.
HRTF handling	SOFA files read with a Python SOFA library	Custom loader	Public datasets in AES69 format.
Room and reverb	Custom image-source and feedback delay network code	Open spatial audio libraries (for example Resonance Audio, Steam Audio)	Open libraries can speed up the first version; check their licenses.
Loudness	A BS.1770 meter library or ffmpeg's loudness filter	Custom meter	Needed for matching and QC.
Encoding	FFmpeg with the FLAC encoder	Native FLAC tools	Lossless output.
Job queue	A simple local queue in Python	Redis queue, a workflow tool	Matches Spotify's batch design.
Storage	Local folders with JSON metadata	A small database	Mirrors the track layout in Chapter 6.
Listener app	Web app using the Web Audio API, or Android	Flutter, React Native	Web is fastest to share.
Native audio on Android	Oboe or AAudio for low latency	Standard media player	Only needed for later real-time work.
Performance code	Rust or C++ for hot loops	Python with NumPy only	Fits your existing DSP and WebAssembly work.

Where your existing work can help

You already have projects in acoustic measurement, calibration and low-level audio code. Reusing them saves time and strengthens the story that the engine is the next step in a body of work.

- Room and headphone calibration ideas can inform the room model and the per-listener fit.
- Your own audio effects can serve as building blocks, such as a compressor in the final bus.
- A web-based, WebAssembly renderer can later run the final stage on the listener's device.

Two implementation paths

	Path 1: Python offline engine plus web player	Path 2: Offline engine plus Android app
What you build first	Batch engine and a browser demo.	Batch engine and a native app.
Sharing the demo	A link and a video. Anyone can listen.	An installable file and a video.
Effort	Lower.	Higher, with native audio code.
Real-time features later	Possible through the browser's audio engine and WebAssembly.	Possible with native audio.
Recommendation	Start here.	Add if the web demo gets attention.

15 Compute, storage and scale

Numbers keep a plan honest. This chapter estimates what the system costs to run, for your laptop and for a real deployment. All figures are rough planning estimates, to be replaced with measurements from your own machine.

Size of one track

```
Raw audio, 24-bit, 44.1 kHz, stereo:
44,100 samples/s * 24 bits * 2 channels = about 2.12 megabits per second
= about 15.9 megabytes per minute
FLAC typically saves a large share of that. A reasonable planning figure is
about 9 to 11 megabytes per minute, so about 35 megabytes for a 3.5 minute song.
```

What that means at catalog scale

Scenario	Calculation	Result
One mode for 1 million tracks	1,000,000 * 35 MB	about 35 terabytes
One mode for 100 million tracks	100,000,000 * 35 MB	about 3.5 petabytes
Four modes for 1 million tracks	4 * 35 TB	about 140 terabytes

The point is not that this is impossible, since a company the size of Spotify handles very large storage. It is that processing the whole catalog in every mode would be wasteful. The opt-in design, an initial focus on a few modes, and the possibility of rendering on demand for rarely played tracks all keep cost sensible. It is also a strong argument for Option B in the long run, where one set of stems serves every mode.

Processing time

Separation is the heaviest step. A modern separation model typically runs several times slower than real time on a laptop CPU and much faster on a GPU. Ensembling and test-time augmentation multiply that. Rendering and analysis are far lighter. You will measure this on your own machine early in the project, because it decides how many test tracks you can process in a day.

Stage	Relative cost	Notes
Analysis	Low	Seconds per track.
Separation	High	Dominant cost. Improves with ensembling at the price of time.
Mix intelligence	Low	Small models and rules.
Rendering (per mode)	Medium	Convolution and reverb. Scales with the number of modes.
Fidelity guard	Low	Several measurements per render.

Fitting the one-to-two-hour goal

At platform scale, the engine runs on many machines in parallel. A single track does not need a long time even on modest hardware; the main constraint is queueing when many releases arrive together, such as on Fridays, when most new music is released. The design handles that with a queue, priority for opted-in releases, and more workers on release days.

> **Working with a laptop and mobile data**
>
> On your laptop, plan work in a way that survives limited power and time: process in small batches, save intermediate results, make every stage resumable, and keep a short list of showcase tracks you re-render as the system improves.

Energy and cost

A serious proposal would include a cost per track. The demo can report a measured figure for processing time per minute of audio.

16 Limitations and how to solve them

You asked for every limitation to be listed, with ways to solve each one without giving up the core idea. That is how this chapter is organised. For each problem, there is the cause, what it would sound like or mean, and the way around it.

Technical limitations

Limitation	Why it happens	How to address it
Imperfect stem separation	Overlapping sounds and effects cannot be perfectly untangled from a mix.	Ensembling, test-time augmentation, residual refinement, fine-tuning. Keep a residual and blend part of the original. Put width on artifact-tolerant elements.
Stereo-only source material	Artists deliver one stereo master. There is no multichannel information to recover.	Treat the work as intelligent upmixing. Be clear in the pitch that it is not a replacement for a hand-made Atmos mix, and measure the gap.
Generic HRTFs	Every ear is different. A generic filter fits some listeners and not others.	Offer a few HRTF profiles and a quick selector. Study personalisation as a later feature.
Front-back confusion and in-head sound	A known weakness of binaural audio without head movement.	Use good early reflections and distance cues. Head tracking in a later design.
Speakers and cars	Both ears hear both speakers, and cabins differ a lot.	Start with headphones. Treat car mode as its own research task and describe how well it generalises.
Bluetooth	Bluetooth cannot carry full lossless, and it adds latency.	Use the highest-quality codec available and say so. Pre-rendered audio does not need low latency, which suits Bluetooth. Real-time head tracking would.
Colouration from filters	HRTF and room processing change the tone.	Use compensation filters, check the downmix, and give a neutral mode.
Loudness changes	Spatial processing alters perceived level.	Match loudness against the original with an objective meter.
Mono compatibility	Wide mixes can cancel when summed to mono.	The guard tests mono sum and rejects cancellation.
Old and mono recordings	Little stereo information and limited bandwidth.	A dedicated gentle profile. Never pretend the recording was wide.
Style classification errors	Genre labels overlap, and many African and fusion styles are under-represented in public data.	Build a diverse labelled set, use soft probabilities rather than hard labels, and blend templates.

Product and business limitations

Limitation	Why it happens	How to address it
Rights and consent	Processing a recording creates a new version. Contracts and approval are needed.	Opt-in only, signed agreement, preview and approval, and an easy way to withdraw.
Artistic intent	Some artists want their stereo image untouched.	Never default on. Respect declines. Offer intensity and protected-element controls.
Perception of authenticity	Listeners may assume a spatial mix was made by the artist.	Label it clearly as an automated spatial mix.
Storage cost	Each mode is another full-size file.	Start with few modes, use opt-in, and move toward rendering from stems on the device later.
Naming and trademarks	Famous studios and format names are protected.	Use original names and neutral descriptions.
Competition from device features	Phones already offer generic upmixing.	Differentiate on per-track intelligence, artist control and testing evidence.
Spotify may not respond	Large companies get many unsolicited ideas.	Make the work self-evidently valuable and easy to try. Aim at being a strong portfolio project whatever the reaction.

Project and personal limitations

Limitation	Why it happens	How to address it
Zero budget	No paid cloud, no paid datasets.	Use open tools, free GPU tiers in small resumable runs, and your own laptop. Choose showcase tracks carefully.
Limited compute	Separation and fine-tuning are heavy.	Process in batches, cache results, test on short clips first and full songs later.
Training data rights	Commercial music cannot be freely used for training.	Use open datasets under their licenses, your own sessions and collaborators' material.
Scope	The idea covers research, engineering, design and legal thinking.	Build in stages with a working system at the end of each one, so progress is never lost.
Subjective results	Taste varies.	Blind tests with enough listeners, and honest reporting.
Study and other commitments	You are a student with other priorities.	Keep the stages small enough to finish, and document as you go.

The honest ceiling

Even with strong engineering, there is a limit to what can be done from a stereo file. A skilled engineer mixing in Atmos has access to every instrument and makes creative choices that no automatic system can fully match. The goal of Stage is not to equal that in every case. The goal is to reach the large majority of the catalog that will never receive an Atmos mix, and to make that experience as good as it can be. A claim of that kind is both more honest and more convincing.

17 Build roadmap

The project is large, so the roadmap is organised as stages. Each stage ends with something that works and can be shown, so the project always has a presentable state. The stages are ordered so the foundations come first and the most ambitious parts come when they have a solid base to stand on. There are no calendar dates, because pace depends on your study schedule.

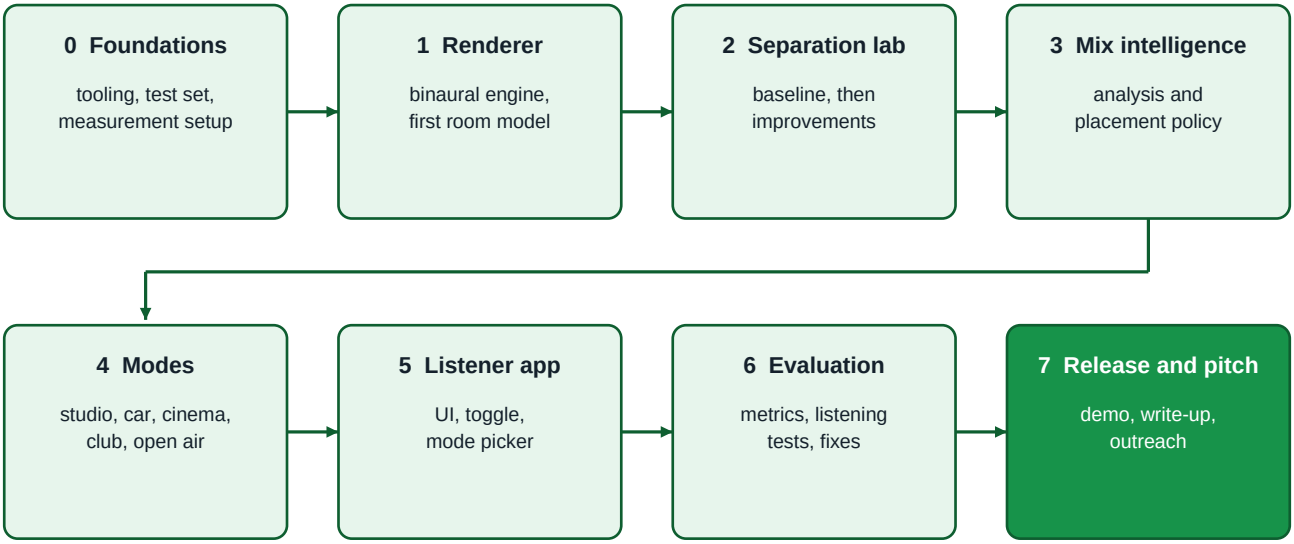


Figure 6. The build stages. Each one produces something that can be played, measured and shown.

Stage by stage

Stage	What you build	Done when
0 Foundations	Environment, a test library across eras and genres, loudness and metric tools, a way to run blind listening tests, and baseline measurements of the separator on your laptop.	You can run any track through a baseline and measure the result.
1 Renderer	A binaural renderer with measured HRTFs, a simple room model, and a command-line tool that places labelled test sounds at chosen positions.	Test sounds appear at the intended positions on headphones, and the result is stored as lossless FLAC.
2 Separation lab	The baseline separator with a residual, then ensembling, test-time augmentation, refinement and fine-tuning where data allows.	Measured improvement over the baseline on held-out tracks, and listening tests agree.
3 Mix intelligence	The analysis stage, the style classifier, the rule-based placement policy for each style template, and the fidelity guard.	Showcase tracks across styles sound good, and the guard passes.
4 Modes	Studio Room, Cinema, Club and Open Air first, then Car as a research task.	Each mode is clearly different and each passes the guard.
5 Listener app	A player with a spatial toggle, a mode picker, an instant A/B switch, and the interface from Chapter 12. A simple mock artist dashboard.	A person can pick up the app and understand the feature in seconds.
6 Evaluation	Listening tests with a real panel, objective results across the full test library, and fixes.	A results report that states clearly where the system works and where it does not.
7 Release and pitch	The demo video, the technical write-up, the audio comparison page and the outreach package.	A package that a stranger can understand and try in a few minutes.

Where to be ambitious

You asked that the plan not be cut down to a minimal version. The roadmap does not do that: it includes improving the separator, style-aware mixing, multiple modes, an artist control design, and a proper evaluation. The ordering simply makes sure that each ambitious piece lands on something solid. Good candidates to push further, once the base is working, are:

- A learned placement policy trained from listening-test preferences.
- A genre-aware separator fine-tuned on African popular music, which is under-represented in public datasets and would be a genuine contribution.
- A real-time, on-device renderer built from your WebAssembly and DSP skills, with head tracking.
- A published open evaluation set and protocol for stereo-to-spatial upmixing, which others could use.

18 Presenting it to Spotify

A great project needs a great way to be seen. People at a large company decide very quickly whether something is worth their time, so the package should make it easy to understand and try.

What the package contains

1. **A one-page summary.** The problem, the idea and a link to the demo.
2. **A short demo video.** Under two minutes, showing a non-Atmos phone, a quick switch between stereo and spatial, and two or three modes.
3. **An audio comparison page.** Blind A/B clips on tracks you have the rights to publish, so people can hear the difference with headphones.
4. **A technical write-up.** Architecture, methods, measurements and limitations. This document is the foundation for it.
5. **The listener app.** A link people can open and try.
6. **Honest results.** What worked, what did not, and what would be needed next.

The message

Lead with the listener benefit and the artist control, then show the evidence. A message that fits your idea well is: "Spotify listeners could hear any opted-in song around them, in modes that suit where they are, with no special hardware and with artists in control. Here is a working prototype and the evidence for how it sounds." Avoid claiming it is better than Apple in general. Say where it is stronger (catalog reach, device independence, artist control) and where Atmos is stronger (hand-crafted mixes).

Where to share it

- A public project page and repository with a clear README.
- A short video on the platforms where audio and engineering people gather.
- Direct messages to engineers and researchers at Spotify who work on audio, personalisation or machine learning. Keep them short and link to the demo.
- Applications to roles and programmes at Spotify and other companies, with the project as a central piece of your portfolio.

> **A realistic expectation**
>
> No one can promise how a company will respond. What you control is the quality of the work and how easy it is to evaluate. A project that is well built, honestly measured and clearly presented is valuable for your path whether or not Spotify replies, and it speaks to the exact skills that machine learning and audio roles look for.

Protecting yourself

- Publish only audio you own or that is openly licensed.
- Do not use Spotify's trademarks or logos as if the project were an official product. Make it clear it is an independent proposal.

A Appendix A: glossary

Term	Meaning
Ambisonics	A way to represent a full sound field around a point, often used as an intermediate format for spatial audio.
Azimuth	The angle of a sound around you in the horizontal plane.
Binaural	Audio produced for headphones so that each ear receives the signal it would hear in a real space.
Bleed	Sound from one instrument that leaks into another instrument's separated track.
Convolution	The mathematical operation that applies a filter, such as a head-related filter or a room response, to a signal.
Crosstalk cancellation	A method for making speakers behave more like headphones by cancelling the sound that reaches the wrong ear.
Dolby Atmos	An object-based audio format in which mix engineers place sounds in three-dimensional space.
FLAC	Free Lossless Audio Codec. It compresses audio without any loss of information.
HRTF and HRIR	Head-related transfer function and its time-domain form, the impulse response. They describe how the head and ears filter sound from a direction.
ILD and ITD	Interaural level difference and interaural time difference: the main cues for direction.
LUFS	A unit of perceived loudness defined by ITU-R BS.1770.
Mid/side	A way to describe stereo as a sum (mid) and difference (side) signal, useful for measuring width.
Object-based audio	Audio described as separate sounds with positions, rendered at playback for the listener's system.
Residual	What is left of the original mix after the separated stems are subtracted.
SDR	Signal-to-distortion ratio, a standard score for the quality of source separation.
SOFA	A file format for storing spatial acoustic data such as HRTFs.
Stem	A separated group of instruments or voices from a mix, such as vocals or drums.
True peak	The highest level a signal reaches after conversion, including peaks between samples.

B Appendix B: sources and further reading

Verified for this plan

- Spotify for Artists help, "Audio file formats for Spotify": support.spotify.com/artists/article/audio-file-formats/
- Spotify lossless launch coverage, September 2025: soundguys.com and audioxpress.com reports on 24-bit/44.1 kHz FLAC for Premium
- Reports on Spotify's backend: engineering summaries from industry blogs (approximate; Spotify's own engineering blog is the primary source)
- Consumer guides confirming that Spotify offers no native Dolby Atmos as of early 2026

Where to read more (search for these by title or author)

- Defossez and colleagues, work on Demucs, and Rouard, Massa and Defossez, "Hybrid Transformers for Music Source Separation".
- The MUSDB18-HQ and MoisesDB dataset papers, and the Music Demixing Challenge reports.
- AES69 (SOFA) specification and the public HRTF datasets: CIPIC, HUTUBS, SADIE II.
- ITU-R BS.1770 and EBU R128 on loudness measurement.
- Woodworth and Schlosberg on the spherical-head model of interaural time difference.
- Introductions to binaural synthesis, the image-source method, and feedback delay network reverberation.
- Documentation for open spatial audio toolkits such as Resonance Audio and Steam Audio.

> **Always verify**
> Details of tools, licenses and platform features change. Before you rely on any of them, check the current documentation and license terms.